

3D Ground-Penetrating Radar Rebar Identifiability: Progress Report

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Executive Summary

The current three-dimensional ground-penetrating-radar simulation campaign has established a controlled testbed for close, finite-length rebar geometry. The central result is not merely that the correct geometry can be recovered. The important result is that the workflow can expose a physically plausible wrong branch, measure how close it is to the truth, and show when source uncertainty changes the ranking.

The current evidence supports five conclusions:

1. The three-dimensional gprMax workflow is operational for finite rebar scenes.
2. Gross wrong geometries, such as a fused single bar or incorrect spacing, are clearly separated from the truth in the tested synthetic cases.
3. Finite bar length is now the nearest geometry ambiguity: the 245 millimeter competitor is consistently closer to the 260 millimeter truth than the 200 or 230 millimeter competitors.
4. Source mismatch is the strongest current nuisance effect. A wrong 245 millimeter geometry with the matched source can look better than the true geometry evaluated with the wrong source.
5. When source frequency is treated explicitly, the true 260 millimeter geometry remains selected on line000, line005, and line010.

This supports a publication framing around controlled three-dimensional identifiability of close multi-rebar geometry with explicit competing branches, confidence margins, source sensitivity, and unresolved-label discipline.

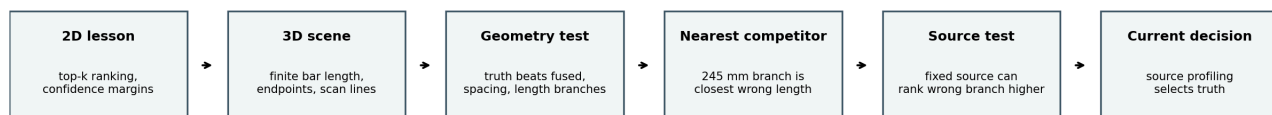
Research Objective

Prior work already exists on rebar detection, localization, diameter estimation, full waveform inversion, radar-array imaging, and neural-network ground-penetrating-radar inversion. The useful contribution here is therefore not "first rebar detection."

The sharper objective is:

Given a controlled three-dimensional close-rebar scene, determine when the true geometry is distinguishable from nearby wrong geometries, and quantify how source uncertainty changes that decision. This objective is a direct extension of the mature two-dimensional work. The two-dimensional campaign established the outer inversion logic: generate a known truth, score many candidate geometries, report the nearest wrong competitor, and label weak or unresolved cases. The three-dimensional campaign keeps that logic but adds finite bar length, endpoints, crossline scan position, and multiple scan lines.

Scientific progression of the current 3D campaign



Definitions Used In This Report

Ground-penetrating radar sends a short electromagnetic pulse into a material and records reflections. In concrete, metal rebars produce strong reflections.

gprMax is an open-source electromagnetic simulator. It is used here to create synthetic radar data for known three-dimensional rebar scenes.

B-scan means the radar image along one scan line. The horizontal axis is scan position, and the vertical axis is time sample.

C-scan means a set of parallel scan lines over an area. The current source study samples three lines from that idea: line000, line005, and line010.

Candidate geometry means one possible explanation for the radar data, such as a 245 millimeter bar-length branch tested against a 260 millimeter truth.

Best-versus-next margin means the difference between the selected best geometry and the nearest distinct wrong geometry. A larger margin means a clearer decision.

Source profiling means allowing source-related nuisance choices, such as source frequency and small timing shifts, before deciding which geometry is best.

Full waveform inversion means fitting simulated waveforms directly to observed waveforms. The current field archive is not ready for calibrated field full waveform inversion.

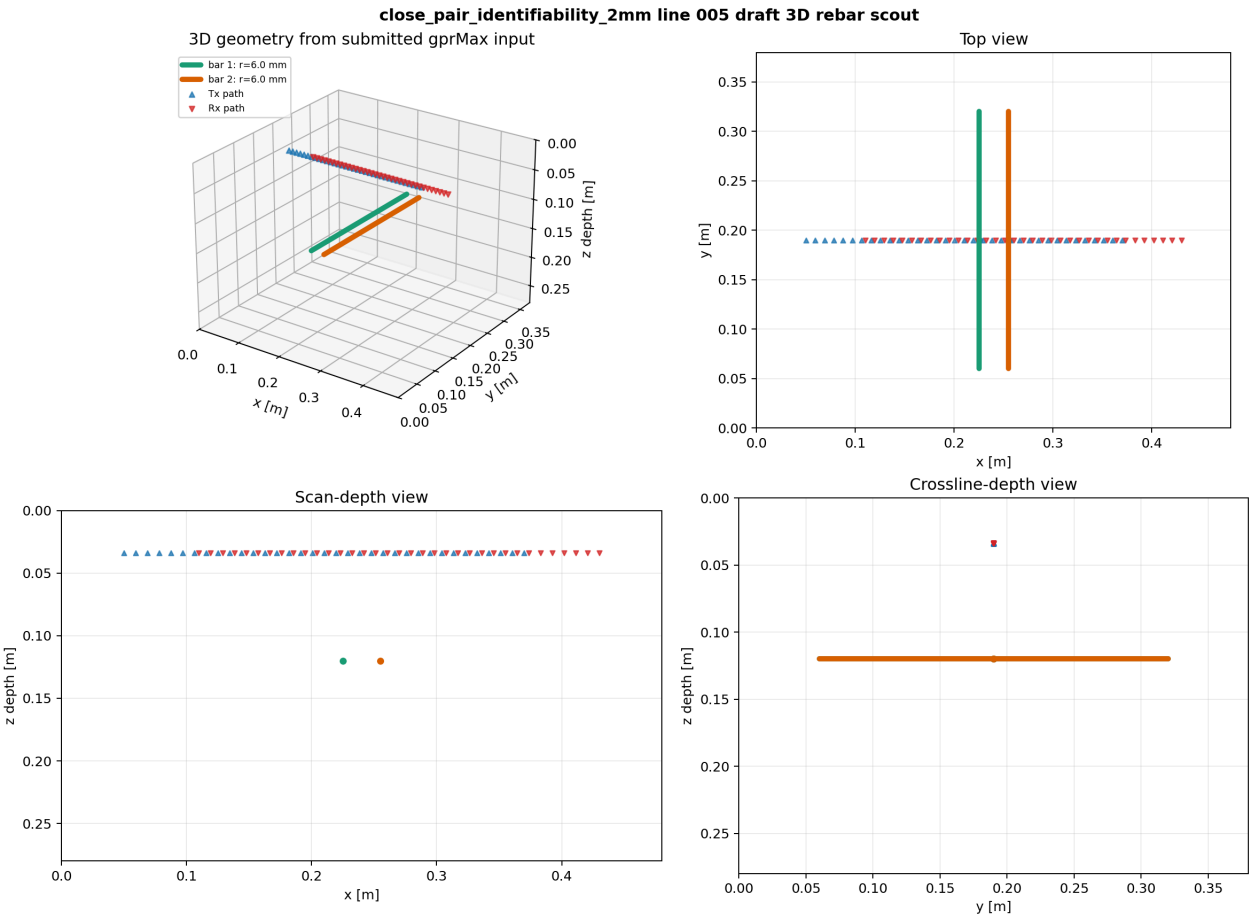
High-performance computing refers to cluster computing on Perlmutter. The current project uses Perlmutter because three-dimensional finite-difference simulations become expensive as candidate grids grow.

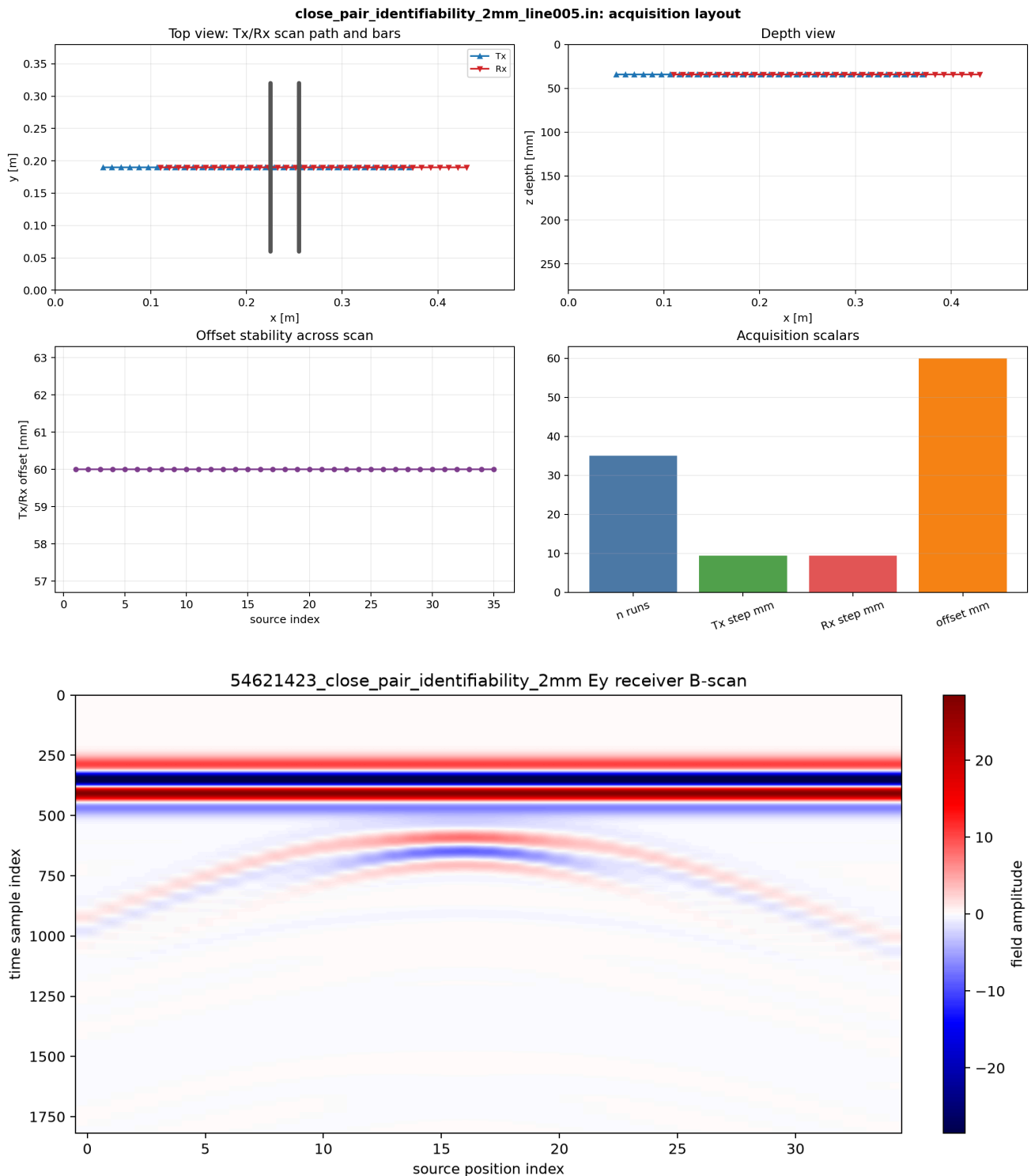
Experimental Setup

The core synthetic scene contains two metal rebars embedded in a concrete-like domain. The current truth branch uses two 260 millimeter bars, 30 millimeter spacing, and 6 millimeter radius. The tested wrong branches include:

- a fused single-bar explanation;
- 20 millimeter and 40 millimeter spacing alternatives;
- 200, 230, and 245 millimeter finite-length alternatives;
- source-frequency alternatives around 1.500 and 1.575 gigahertz.

The compact case names encode those choices. For example, `sep30_len245_r6_line005` means a two-bar candidate with 30 millimeter spacing, 245 millimeter bar length, 6 millimeter radius, evaluated on scan line 005.

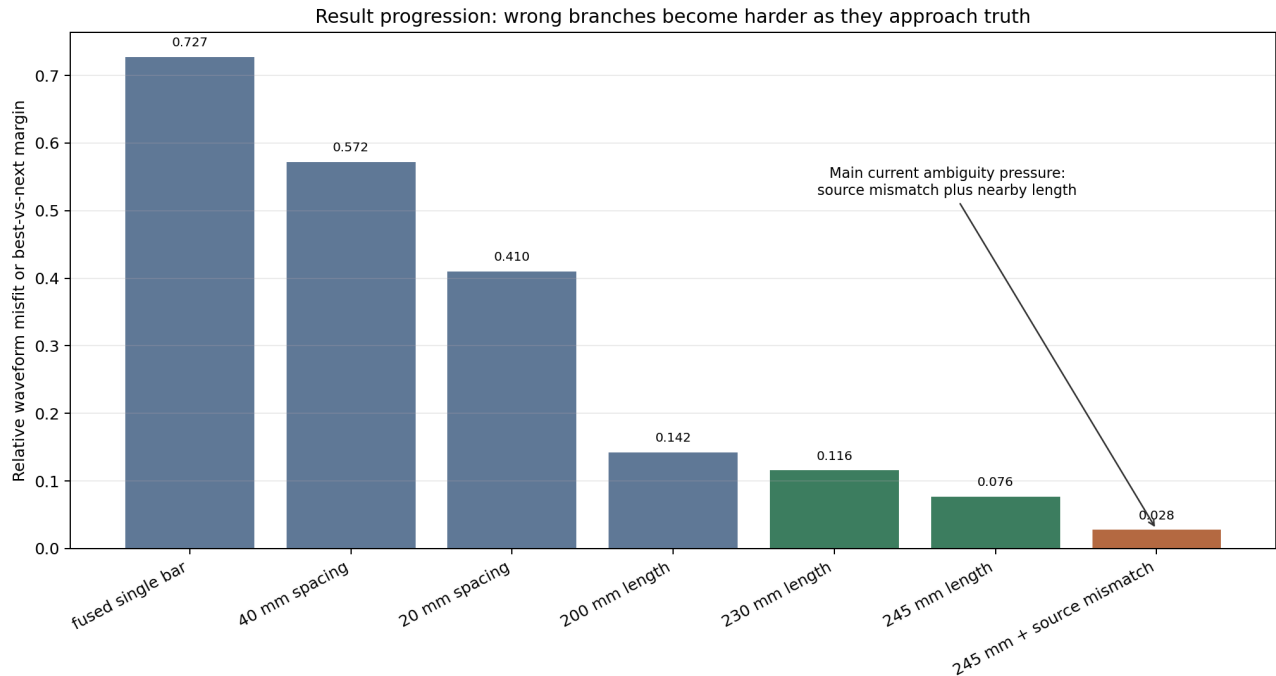




Result 1: The Main Ambiguity Has Narrowed To Finite Length

The early three-dimensional tests rejected coarse wrong explanations. A fused single bar, 20 millimeter spacing, and 40 millimeter spacing all remained far from the true branch. The harder cases are not those gross geometry errors. The harder cases are wrong finite lengths that preserve the same spacing and radius.

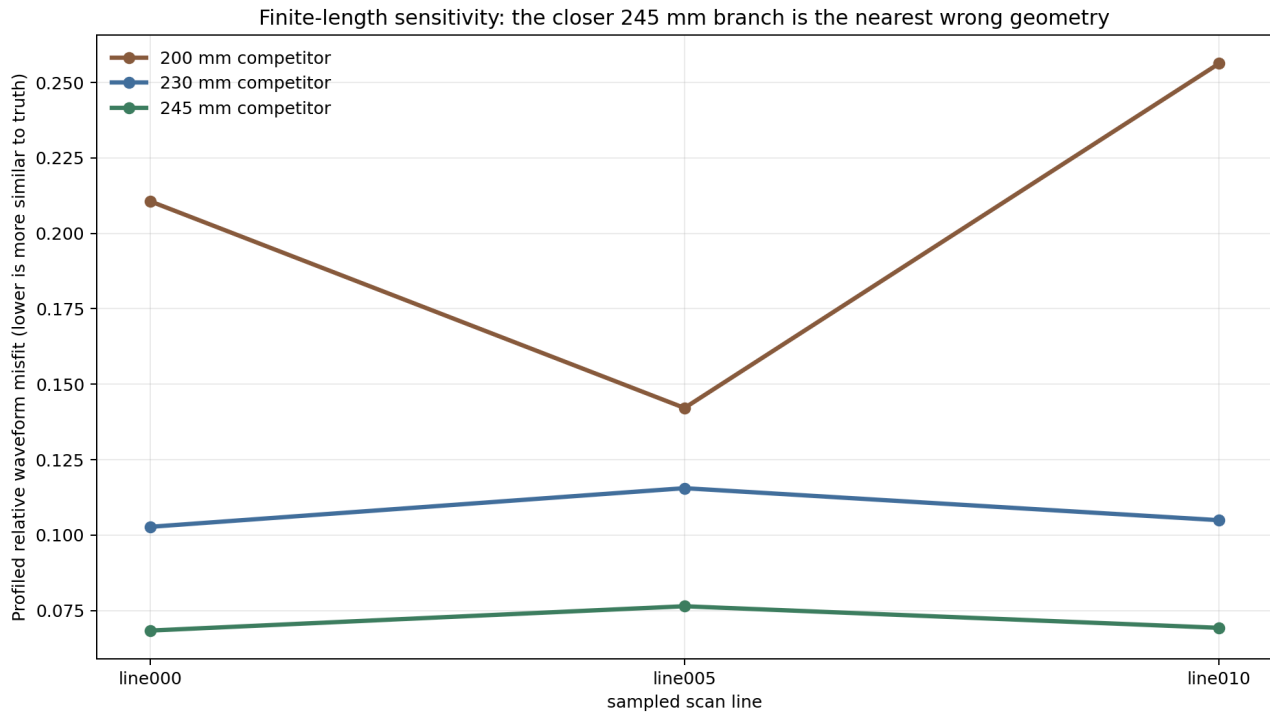
stage	evidence	implication
3D forward model established	The close-pair gprMax scene ran and produced interpretable B-scan outputs.	The project can now test finite-length three-dimensional rebar hypotheses directly.
Wrong spacing and fused-bar alternatives rejected	Fused, 20 mm spacing, and 40 mm spacing branches are much farther from the truth than the true branch.	The current ambiguity is not simply one-bar versus two-bar or gross spacing confusion.
Finite length is the nearest geometry ambiguity	The 245 mm branch is consistently closer than the 200 mm and 230 mm branches across the sampled lines.	Length sensitivity is real and should be treated as a primary 3D parameter.



Result 2: The 245 Millimeter Branch Is The Nearest Length Competitor

Across the sampled scan lines, the 245 millimeter competitor is consistently closer to the 260 millimeter truth than the 200 or 230 millimeter competitors. This is the expected but important result: as the wrong length approaches the true length, the waveform becomes harder to distinguish.

competitor length	line000	line005	line010	interpretation
200 mm	0.2107	0.1421	0.2563	short branch, still separated
230 mm	0.1027	0.1155	0.1049	intermediate competitor
245 mm	0.0683	0.0764	0.0693	nearest finite-length competitor



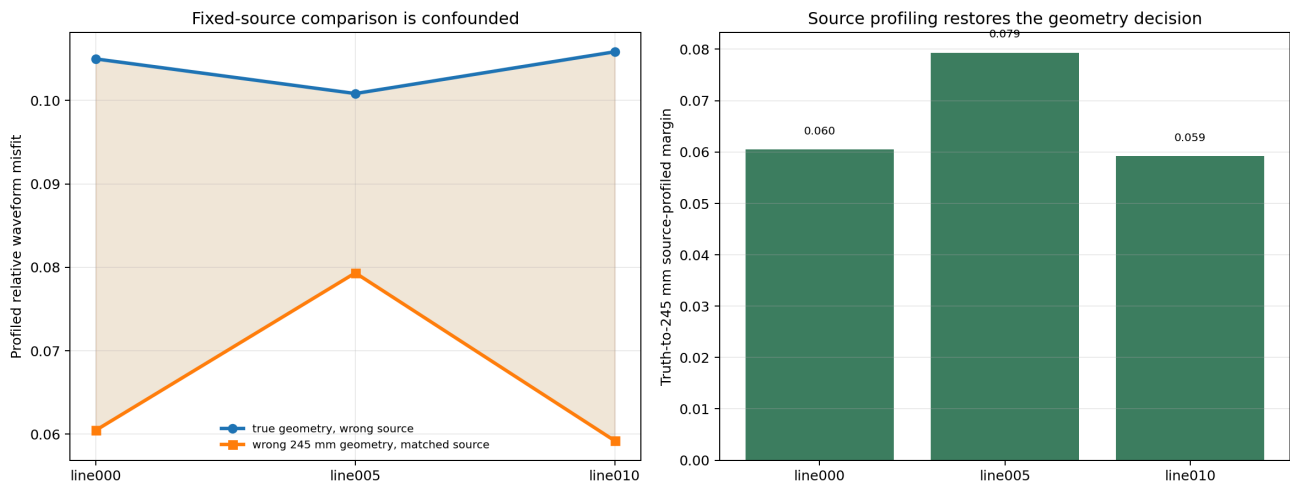
Result 3: Source Mismatch Can Change The Apparent Winner

The most important source-sensitivity result is that geometry and source cannot be interpreted independently. On all three sampled scan lines, the wrong 245 millimeter geometry with the matched 1.575 gigahertz source has lower misfit than the true 260 millimeter geometry evaluated with the nominal 1.500 gigahertz source.

line	wrong 245 mm, matched source	truth, wrong source	fixed-source confounding gap	truth selected after source profiling
line000	0.0605	0.1050	0.0445	True
line005	0.0793	0.1008	0.0215	True
line010	0.0592	0.1059	0.0467	True

This does not mean the current geometry result fails. It means fixed-source scoring is unsafe as the final decision rule. Once the source is profiled explicitly, the true 260 millimeter geometry remains selected on all three lines.

mean fixed-source confounding gap: 0.037571508728419266
mean source-profiled truth margin: 0.06632340960632299
minimum source-profiled truth margin: 0.059189017225787774



Interpretation

The current three-dimensional evidence supports a focused identifiability claim. The tested data do not merely recover a known truth; they show how wrong branches become competitive and where nuisance parameters matter.

The practical interpretation is:

- The tested close-pair geometry is resolved against the current wrong-geometry set.
- The nearest geometry competitor is finite-length, especially the 245 millimeter branch.
- Source uncertainty can make a wrong branch look better than the true branch if the true branch is evaluated with the wrong source.
- Source profiling is therefore not optional bookkeeping; it is part of the scientific result.

stage	evidence	implication
Source mismatch is the main current nuisance	At all three sampled lines, 245 mm with the matched 1.575 GHz source beats truth evaluated at 1.500 GHz.	A fixed-source inversion can overstate confidence or select the wrong branch.
Source profiling preserves truth for the current case	After source profiling, truth remains selected on line000, line005, and line010.	The result is publishable as a controlled identifiability and nuisance-sensitivity study, not as a broad field claim.

Perlmutter Readiness And Compute Boundary

Perlmutter is being used because three-dimensional finite-difference simulations become expensive when each candidate geometry requires many source positions and output traces. The current environment evidence is sufficient for a guarded next check, but it does not authorize a broad compute campaign.

visible GPU inventory: GPU 0: NVIDIA A100-PCIE-40GB (UUID omitted)
gprMax version: 3.1.7
PyCUDA version: 2026.1
PyTorch module smoke: pytorch/2.11.0 / CUDA available = True
current submit gate:
readiness_dashboard_evidence_ready_wait_for_recheck_window
submit currently allowed: False
test-only window decision: sbatch_test_only_start_too_far_defer
The next Perlmutter action should remain exactly one guarded full-node pilot only after the existing submit gates turn positive. Scaling before that pilot is inspected would weaken the evidence chain.

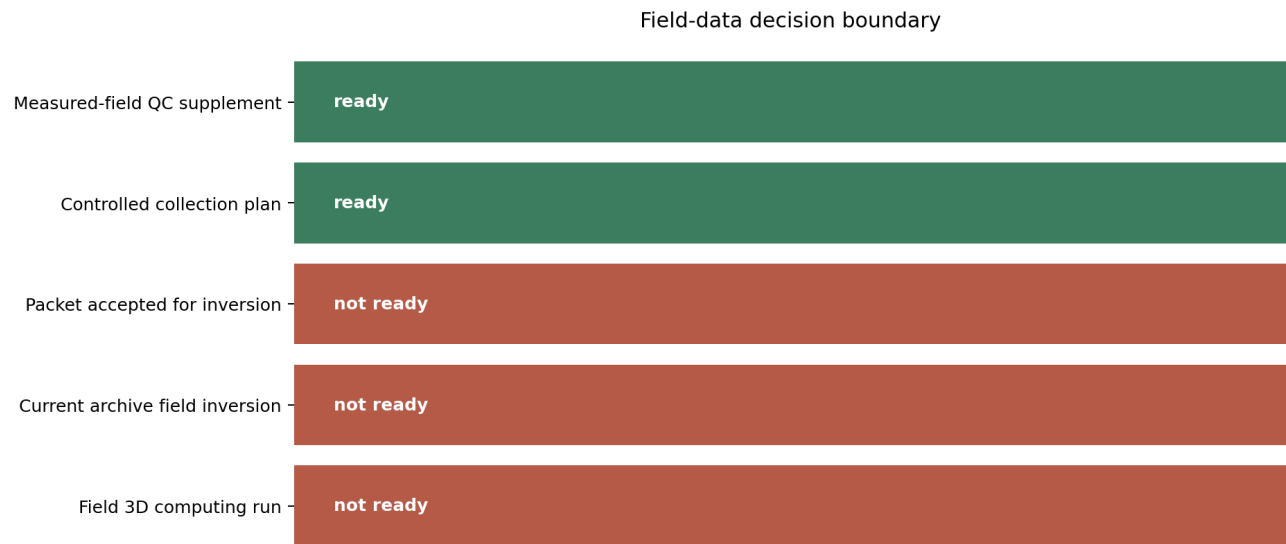
Field-Data Boundary

The measured GSSI 51600S field archive is useful as quality-control evidence, but it is not yet a calibrated field full-waveform-inversion workload and it is not a field three-dimensional computing input.

Current field decision:

field geometry type: independent_2d_line_profiles
ready for controlled collection execution: True
ready for packet acceptance: False
ready for current field full waveform inversion: False
ready for field 3D high-performance computing: False
critical new-data groups:
target_truth_geometry;time_zero_reference;amplitude_reference;profile_target_ge

The next field step is controlled two-dimensional collection with target truth, time-zero references, amplitude references, surveyed profile geometry, and repeat scans. That field path should remain separate from the synthetic three-dimensional Perlmutter path.



Supported Claims And Current Limitations

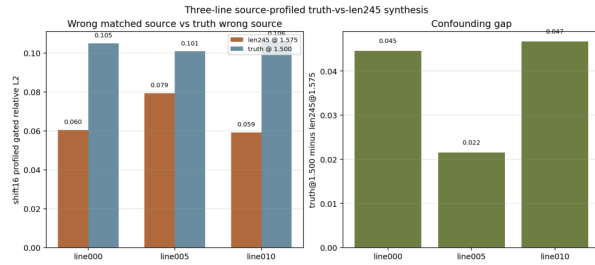
claim	status	limitation
In the current 3D close-pair scout, fixed nominal-source ranking is source-confounded across line000, line005, and line010.	supported_small_batch	Only one perturbed frequency, one nearest finite-length wrong branch, one close-pair geometry, and three y-lines.
Explicit source profiling still resolves the true geometry for truth versus len245 in the current three-line sample.	supported_small_batch	Truth row is self-reference; broader source nuisance and noise studies are not yet included.
The dominant measured source-confounded finite-length competitor is len245, not len230.	supported_center_line_only	Endpoint len230 rows have not been simulated at 1.575 GHz.
More GPU work should be gated by a specific ambiguity question rather than by table completion.	planning_decision	A reviewer-facing study will still need noise, acquisition, and maybe field-data bridges later.

The current study should not claim field performance, dense C-scan generalization, noise robustness, arbitrary rebar orientation, or neural-network integration. The strongest claim is narrower and cleaner: source-aware branch-ranking can expose and resolve the current close-pair finite-length ambiguity in a controlled three-dimensional synthetic setting.

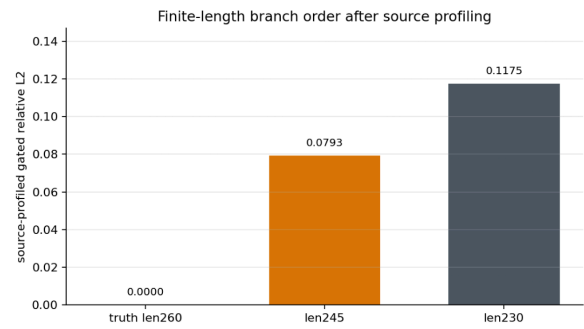
Next Scientific Plan

1. Preserve the current source-confounding result as the first paper-facing three-dimensional finding.
2. Run exactly one full-node Perlmutter pilot only after the current scheduler and submit gates allow it.
3. Inspect that pilot's logs, HDF5 outputs, figures, and branch rankings before any scaling.
4. Use the next compute block to answer a specific ambiguity question, not to fill a table mechanically. The highest-value candidates are dense y-line sampling around the 245 millimeter branch, explicit source/noise sensitivity, and acquisition-layout sensitivity.
5. Keep field-data work separate until the controlled collection packet passes its acceptance gates.
6. Treat 3DInvNet as a later baseline or comparison, not the current main contribution.

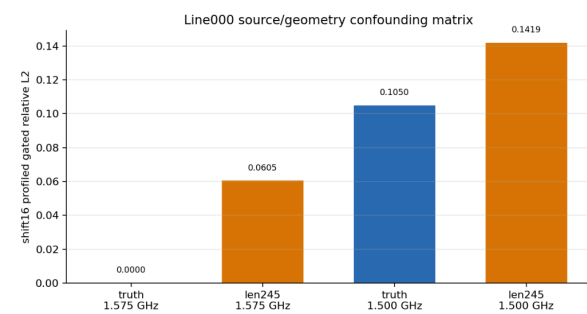
A. Three-line source-confounding summary



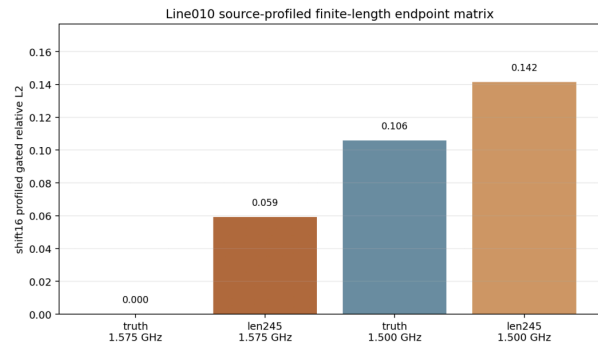
B. Line005 finite-length source-profiled order



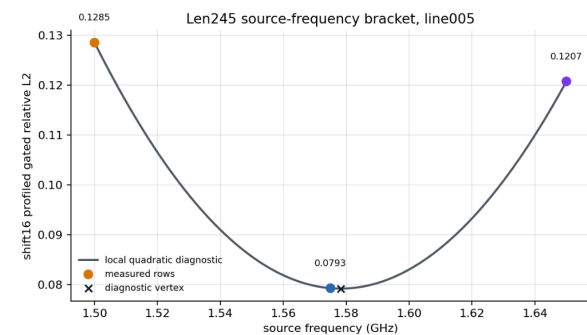
C. Line000 endpoint matrix



D. Line010 endpoint matrix



E. Line005 len245 source-frequency bracket



Current evidence scope

Source-aware ranking currently selects the true 260 mm finite-bar geometry over the nearest 245 mm finite-length branch on line000, line005, and line010. This supports a controlled identifiability result for the simulated close-pair case.

Evidence still outside the current result set

- field-data performance
- broad noise-robustness statistics
- dense C-scan generalization
- neural-network integration
- first-in-field rebar sizing or detection claims

Compute status: this panel summarizes existing results only and does not include a new submitted GPU job.

Conclusion

The current three-dimensional campaign establishes a controlled finite-length rebar identifiability workflow, not merely a larger version of the earlier two-dimensional simulations. The first result is that coarse wrong geometries are separated from the truth, while the meaningful remaining ambiguity is narrower: nearby finite-length branches and source uncertainty.

The closest tested wrong geometry is the 245 millimeter branch against the 260 millimeter truth. Fixed-source scoring can make this wrong branch look artificially competitive, but source-profiled ranking still selects the truth on the sampled scan lines. This makes source treatment part of the scientific result rather than a secondary implementation detail.

These results support a focused publication path around acquisition-aware three-dimensional identifiability of close multi-rebar geometry, with explicit competing branches, confidence margins, source sensitivity, and unresolved labels. The next stage should extend this result through a small Perlmutter pilot with denser scan-line sampling and controlled source/noise variation before any broader compute campaign is justified.